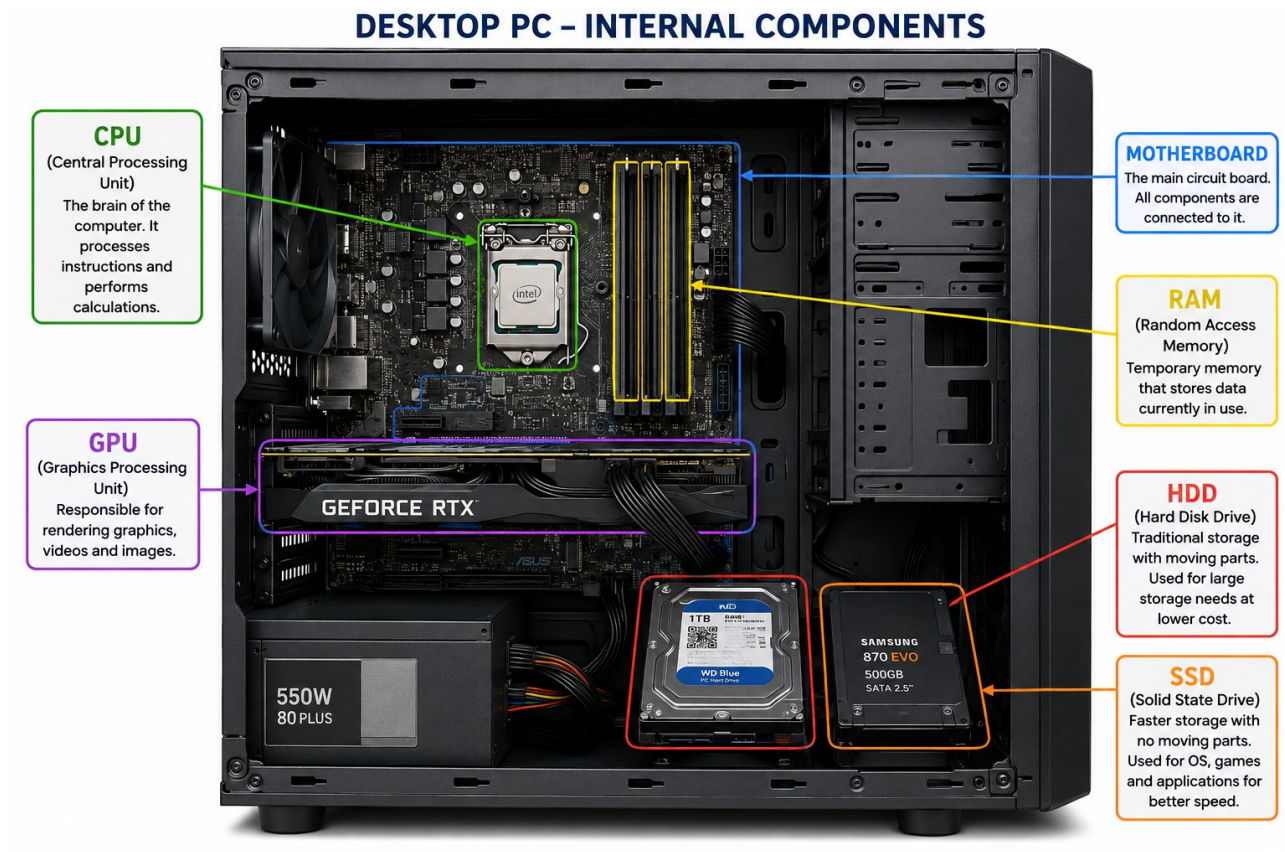


1. Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Speed	Slower	Much Faster
Data Access Time	Higher	Very Low
Durability	Less durable (moving parts)	More durable (no moving parts)
Shock Resistance	Low	High
Noise	Produces noise	Silent
Power Consumption	Higher	Lower
Storage Capacity	Usually larger for lower cost	Usually smaller for same cost
Price	Cheaper	More expensive
Typical Use Cases	Desktop storage, backups, large files	Gaming PCs, laptops, operating systems

3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

CPU (Central Processing Unit)

The CPU acts as the brain of the computer. In games like PUBG or Free Fire, it processes player actions, game logic, AI behavior, and communication between hardware components. A faster CPU helps games run more smoothly.

RAM (Random Access Memory)

RAM temporarily stores game data that is currently being used. More RAM allows the game to load maps, textures, and player information quickly without lag. Insufficient RAM can cause slow performance and stuttering.

GPU (Graphics Processing Unit)

The GPU is responsible for rendering graphics, animations, and visual effects. In games like PUBG or Free Fire, it creates detailed environments, character models, and smooth frame rates. A powerful GPU improves graphics quality and gaming performance.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

Component	Why Upgrade?
GPU (Graphics Card)	Improves graphics quality, frame rate, and overall gaming performance.
CPU	Handles game calculations, AI, and physics more efficiently.
RAM	Increasing RAM (e.g., 16GB or more) helps reduce lag and improves multitasking.
SSD	Faster game loading times, quicker boot speed, and smoother game updates.
Power Supply (PSU)	May be needed if installing a more powerful GPU or CPU.
Cooling System	Keeps upgraded components cool and prevents overheating during long gaming sessions.